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NetPractice (Subnetting)

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Hello Again! Now that you have a handle on binary and TCP/IP addressing, we're going to discuss in this article how to take one network address and break it into multiple networks called subnets!

A good question will be what's the purpose of subnetting ?

The purpose of subnetting is to ***take one address range*** and break it down into ***multiple addresses ranges*** so that you can assign each address range to a separate network (subnet) in your network.

We will walk through a subnetting example using Level 3 (don't worry it's easy) :

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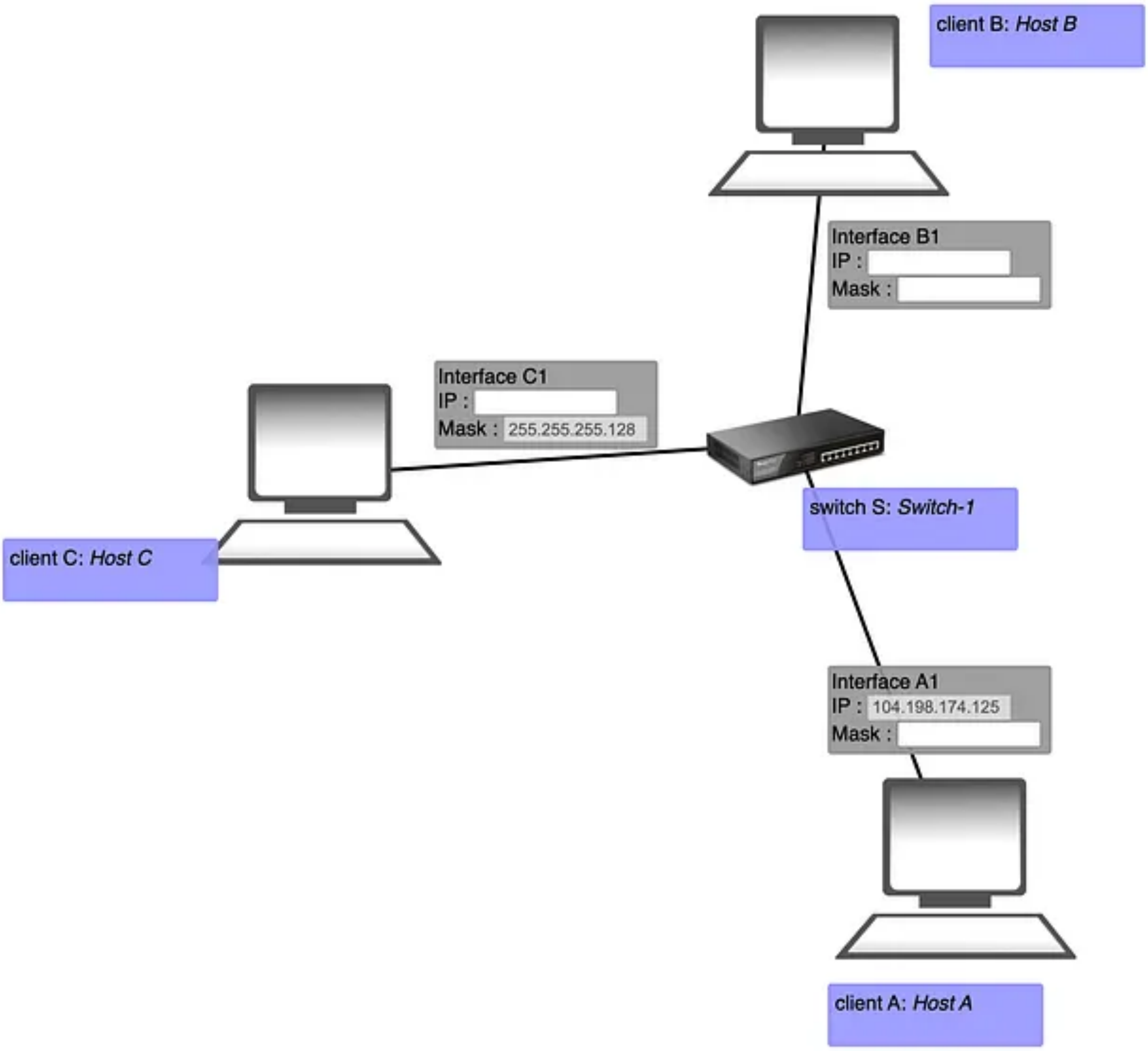

To begin we need to know how many host bits to take to make the desired number of networks. in our example, we want to divide a class A address of 104.198.140.n into 2 networks (remember that our goal is to have 2 networks).

How to Accomplish Subnetting we can accomplish subnetting by manipulating the subnet mask as described in the following subsections.

Step 1. Find the Number of Borrowed Host Bits:

To begin we need to know how many host bits to take to make the desired number of networks. in our example, we want to divide the address of

104.198.140.125 into x networks,

The equations down bellow show how to calculate the number of subnets created per subnet:

$$\textbf{Number of subnets created} = 2^{x}$$

x = Nbr of bits borrowed from the host bits

The first thing we need to know how many bits stolen or masked from the host ID portion, in our case we've masked 1 bit to do the subnetting so we come up with

First Octet	Second Octet	Third Octet	Fourth Octet
255	255	255	128
11111111	11111111	11111111	10000000

25 bits for the Network ID (in Red) , 7 bits for the Host ID (in Green)

we've decided that 2 bits were needed to mask the subnet mask to do the subnetting so we come up (using our equation) :

The number of subnets = 2 ^ 1 = 2 Subnets

Step 2. Derive the Subnets:

We've specified 2 new subnets, each subnet has its own network address, The network addresses are used to route data packets to the correct subnet.

We know that the subnet mask is 255.255.255.128, so we can start by calculating the Network ID of each of the 2 subnets.

Note : You should be aware that there are four of informations you should know about each network when your calculations are over. These 4pieces of information are :

- Network ID
- First Valid Address
- Broadcast Address
- last Valid Address

To determine each of the listed informations previously, we need to determine all on/off stated of the number of buts that we have stolen.

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In our example 1 bits stoled to create more networks, so there are 2 possible on/off states with the 1 bit : 0,1 , it would look like the following table :

First Octet (Decimal)	Second Octet (Decimal)	Third Octet (Decimal)	Fourth Octet (Binary)
104	198	174	0
-	-	-	10000000
-	-	-	00000000

The next thing to do after calculating all of the on/off state combinations of the 1 bit is to add in the remaining 0s to the bits that represent the host ID portion.

Remember that the original network ID was 104.198.140.125, so the first. second and the third octet will start respectively with 104.198.140 no matter what you change in the binary, because you are starting your work with the third octet.

1. Network ID:

First Octet (Decimal)	Second Octet (Decimal)	Third Octet (Decimal)	Fourth Octet (Binary)	Calculation
104	198	174	0	-
-	-	-	10000000	104.198.174.128
-	-	-	00000000	104.198.174.0

To calculate the network ID of each of the two networks simply we'll be leaving all host bits set to 0 (the non red bits) as shown in the table above,

In this example, there will be 2 network IDs of :

- 104.198.174.128
- 104.198.174.0

2. First Valid Address:

The next number, which can be calculated easily, is the first valid address that can be assigned to a host on each of these networks. To calculate the first valid address, you simply enable the lowest-order bit. The lowest-order bit will be the bit on the far right side.

First Octet (Decimal)	Second Octet (Decimal)	Third Octet (Decimal)	Fourth Octet (Binary)	Calculation
104	198	174	0	-
-	-	-	10000001	104.198.174.129
-	-	-	00000001	104.198.174.1

3. Broadcast Address

The broadcast address is the address that any system will send data to in order to ensure that each system on the network reads the data. To calculate the broadcast address, you will enable all of the host bits and get the outcome in the following table:

First Octet (Decimal)	Second Octet (Decimal)	Third Octet (Decimal)	Fourth Octet (Binary)	Calculation
104	198	174	0	-
-	-	-	11111111	104.198.174.255
-	-	-	01111111	104.198.174.127

4. Last Valid Address

The only additional information you need is the last valid address that may be assigned to hosts on each subnet. To calculate the last valid host address of each subnet, simply subtract one from the broadcast address by disabling the low-order bit (the far right-most host bit).

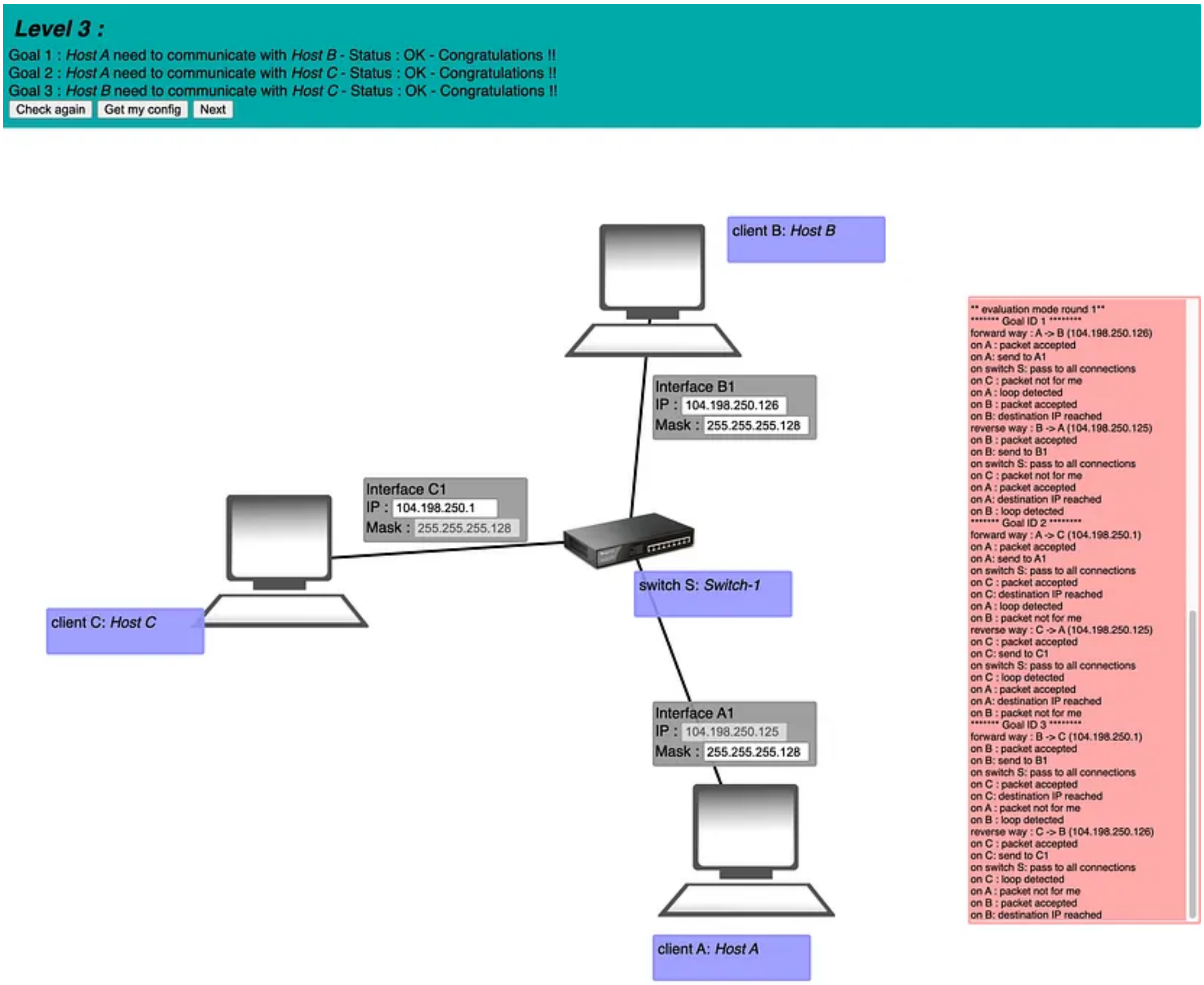
First Octet (Decimal)	Second Octet (Decimal)	Third Octet (Decimal)	Fourth Octet (Binary)	Calculation
104	198	174	0	-
-	-	-	11111110	104.198.174.254
-	-	-	01111110	104.198.174.126

You have now calculated all of the informations required to configure the two network segments that you have created.

The following table summarizes the configuration for each of the two network segments:

####	Network ID	First Valid Address	Last Valid Address	Broadcast Address	Subnet Mask
Subnet 1	104.198.174.0	104.198.174.1	104.198.174.126	104.198.174.127	255.255.255.128
Subnet 2	104.198.174.128	104.198.174.129	104.198.174.254	104.198.174.255	255.255.255.128

Finally let’s check displays how these two network segments will be configured :



Another way to subnetting called magic number :

> **To be continued つづく**

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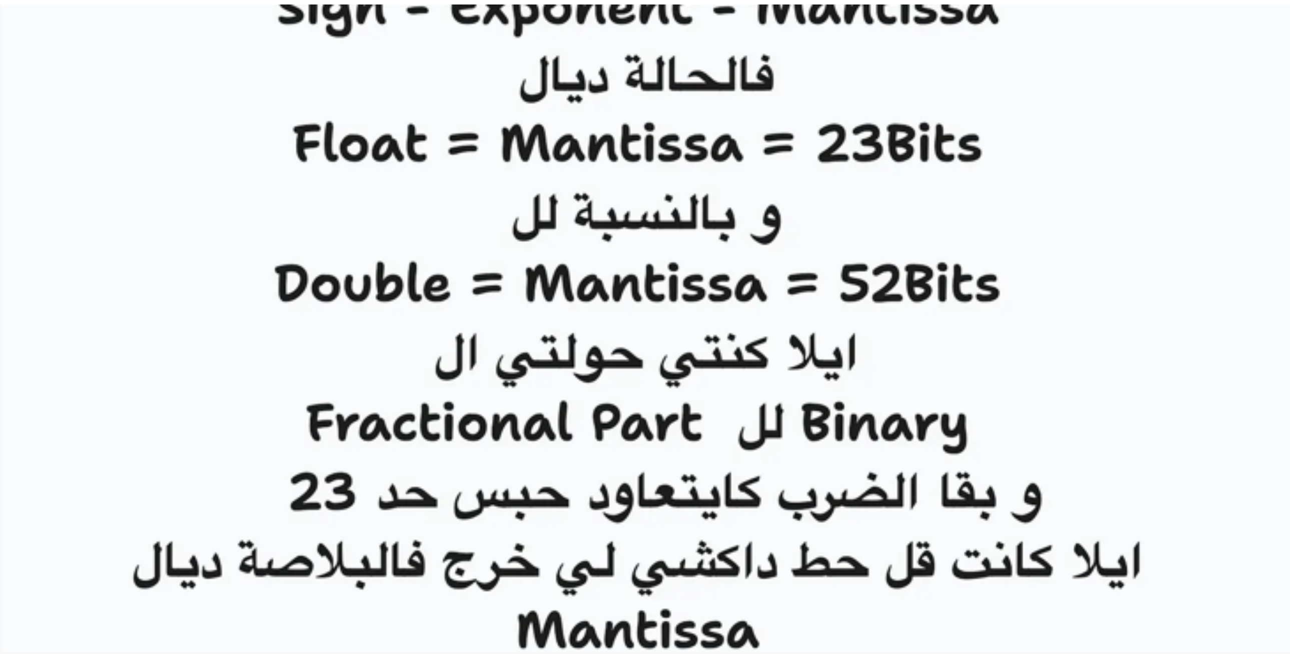

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To represent the decimal number 172.625 in floating point representation, we will use the IEEE 754 format. This format includes 3 main…

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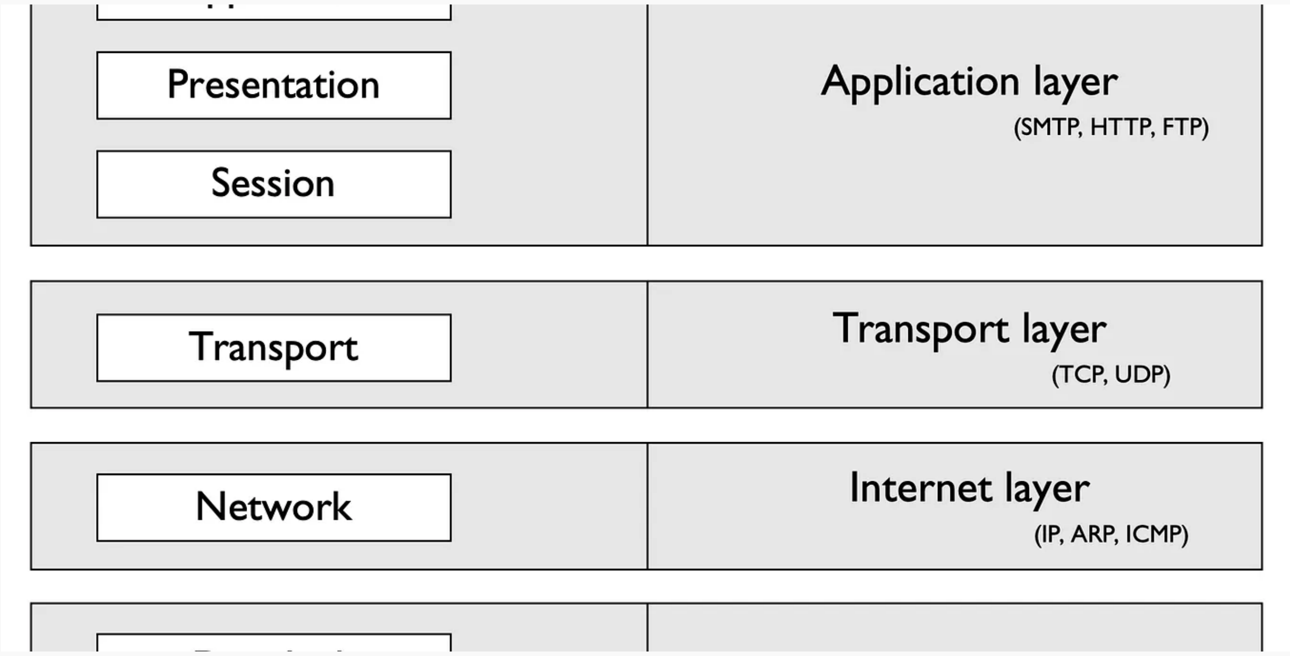

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